

Problem A. Priority Queue

Time Limit 2000 ms
Mem Limit 131072 kB
OS Linux

A priority queue is a data structure which maintains a set S of elements, each of with an associated value (key), and supports the following operations:

- $insert(S, k)$: insert an element k into the set S
- $extractMax(S)$: remove and return the element of S with the largest key

Write a program which performs the $insert(S, k)$ and $extractMax(S)$ operations to a priority queue S . The priority queue manages a set of integers, which are also keys for the priority.

Input

Multiple operations to the priority queue S are given. Each operation is given by "insert k ", "extract" or "end" in a line. Here, k represents an integer element to be inserted to the priority queue.

The input ends with "end" operation.

Output

For each "extract" operation, print the element extracted from the priority queue S in a line.

Constraints

- The number of operations $\leq 2,000,000$
- $0 \leq k \leq 2,000,000,000$

Input	Output
insert 8 insert 2 extract insert 10 extract insert 11 extract extract end	8 10 11 2